

Hi I'm Sameer.

I loved playing tennis for a long time — and I wanted to get better.

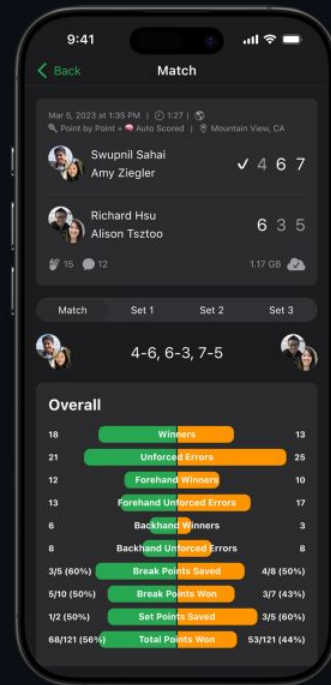


I wanted to analyze the gaps in my tennis game...

Is it fatigue, or is it technique?



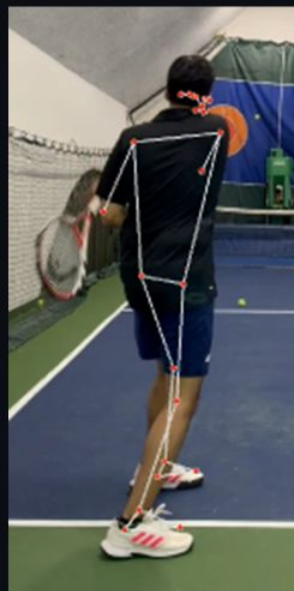
Heart Rate
HRV
Sleep Score



Tennis Stats

The Vision: A tennis prediction engine that uses physiological inputs to develop my in-game tennis strategy

Google MediaPipe



Weak power

Peak racket-hand speed	:	1.15	(power)
Explosiveness into contact	:	0.40	(sharp accel)
Contact height (0-1)	:	0.41	(higher = earlier/cleaner)

Initial Findings: Readiness Score Correlates with Swing Speed

	Readiness	Sleep	HRV	Swing speed
● Fresh days	85	86	98	10.7
● Mid days	74	65	104	9.3
● Tired day	73	68	87	7.3

Readiness → swing speed

$$r = +0.84$$

more recovered → measurably more powerful swing

5 real sessions - exploratory - both numbers independently measured (ring + video)

readiness (x) vs swing speed (y)



Where Do I take it From Here?

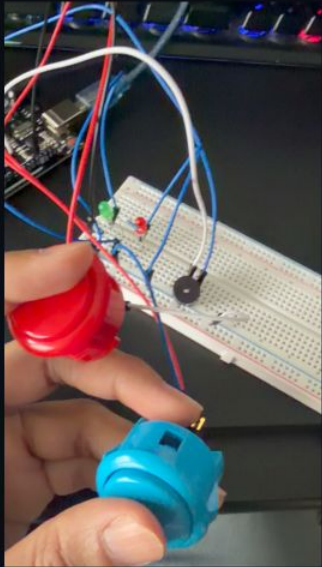
1. Amass a larger dataset – both longitudinally on $n=1$ (me) and with others. Determine which variables tell a story
2. Build the prediction model (e.g., Bad Readiness = Shorter Shots)
3. Build the strategy engine (e.g., Shorter Shots = go to the net more)

But What's the Point...Isn't Everyone Different?

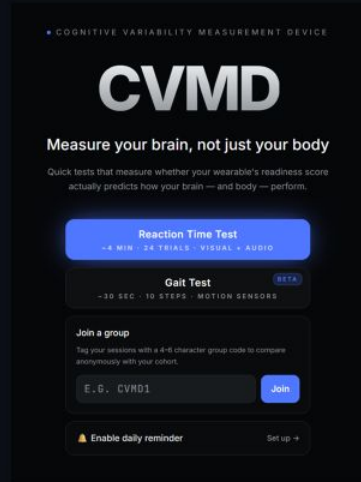
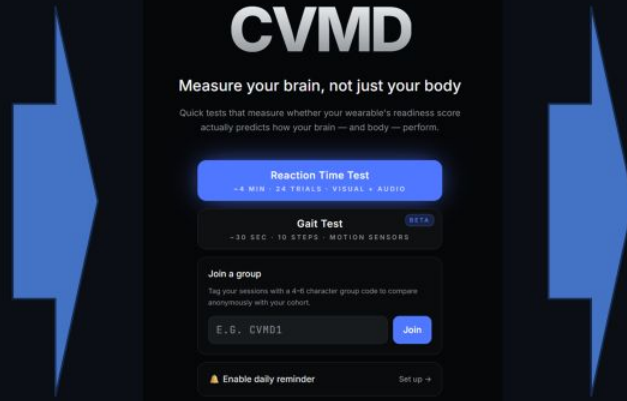
Maybe. But I want to see if there are commonalities in the data that can help me (and hopefully others) know how to use their **body's data** to help play more **strategically**

Did I spend the whole Fellowship working on this?

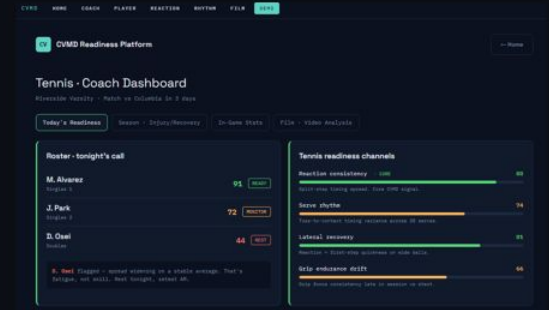
No! I pivoted many times and only started this tennis idea 2 weeks ago...



Original Idea: Hardware Device to better test cognitive impairment through reaction time testing



Next Idea: Software version of test



Next Idea: Coaching dashboard combining player tracking/system of record with reaction time testing for players

Now What?

I will continue to collect data on myself and others to refine the models

Thank You!

You All for Listening to My Pitch

My Fellow Cohort

Michael and the WWSF Team